

Section 552a(b)(3) and may be disclosed to other federal agencies. The "Blanket Routine Uses" apply.
DISCLOSURE: VOLUNTARY. However, nondisclosure may result in missing and/or inaccurate documents in the individuals master personnel records which may result in negative career impacting consequences.
SORN: F036 AF PC A, Effectiveness/Performance Reporting Records

I. IDENTIFICATION DATA (Read AFI 36-2406 carefully before filling in any item)

1. NAME (Last, First, Middle Initial) FIRTH, JORDAN A.	2. SSN 246-61-3834	3. RANK Maj	4. DUTY AFSC 92S0
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5. ORGANIZATION, COMMAND, AND LOCATION

Air Force Institute of Technology (AETC), Wright-Patterson AFB OH

6. PERIOD OF REPORT FROM: 17 Feb 2018	THRU: 24 Aug 2019	7. LENGTH OF COURSE 156 WEEK(S)	8. REASON FOR REPORT ☐ ANNUAL ☒ FINAL ☐ DIRECTED
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9. NAME AND LOCATION OF SCHOOL OR INSTITUTION

North Carolina State University (NCSU), Raleigh NC

10. NAME OR TITLE OF COURSE

Degree Program: PhD

II. REPORT DATA (Complete as applicable for final report)

1. AFSC/AERO RATING/DEGREE AWARDED Doctor of Philosophy	2. ☐ COURSE NOT COMPLETED (List reason in Item 4 below)
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3. DISTINGUISHED GRADUATE ☐ YES (List criteria in Item 4 below)	☒ NO DG PROGRAM
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4. DG AWARD CRITERIA/COURSE NONCOMPLETION REASON

III. COMMENTS (Mandatory)

ACADEMIC/TRAINING ACCOMPLISHMENTS

- **DEGREE AWARDED: Doctor of Philosophy in Aerospace Engineering**
- **FOCUS AREA: Deployable Spacecraft Structures**
- **GPA: 3.89 on a 4.0 scale; successfully completed all PhD requirements**
- **DISSERTATION: "Deployment Mechanisms for Rollable Spacecraft Booms"**
 - Designed 2 separate mechanisms to consistently and reliably deploy booms--advanced field of space structures
 - Validated in-space performance via two manned flights--proved devices will enable lighter cheaper satellites
 - Research recognized by NASA as game changing development--awarded \$180K grant for development/flight test
 - Led 14-mbr team on multi-disciplinary test program culminating in safe, successful flights & data for 4 papers
 - "Advanced Dual-Pull Mechanism for Deployable Spacecraft Booms" published in Journal of Spacecraft & Rockets

PROFESSIONAL QUALITIES (Bearing, appearance, conduct, fitness)

Ratee is geographically separated from evaluator.

- **AFIT Liaison Officer for 6 NCSU officers--mentored to degree completion & strengthened relationships with ROTC**
- **Outstanding communicator; selected as guest lecturer for AFROTC on space exploration and 2 engineering classes**
- **Collaborated with CE dept to validate bridge repair--will be used by NC Trans Dept to repair 226 damaged bridges**
- **Long-term mentor to 6 undergrad engineering students--comm & problem-solving skills will prepare them for jobs**
- **Ran dept-wide equip training--created documentation, ID'd shortfall, oversaw \$15K upgrade restoring key test ability**

OTHER COMMENTS

- **Article titled "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms: Ground Test" accepted for publication in the Journal of Spacecraft and Rockets**
- **Presented paper titled "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms" at the American Institute of Aeronautics and Astronautics SciTech Forum**

IV. EVALUATOR